

Total No. of Questions : 8]

SEAT No. :

PE2434

[Total No. of Pages : 2

[6584]-384

**B.E. (Artificial Intelligence & Data Science)
DATA MODELING AND VISUALIZATION
(2019 Pattern) (Semester - VII) (417522)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

- Q1)** a) What is Computational Statistics, and how does it relate to Data Visualization? **[6]**
- b) How do Presentation Graphics differ from Exploratory Graphics? Provide examples. **[6]**
- c) Discuss the role of Computing in Graphics and its impact on Data Visualization. **[6]**

OR

- Q2)** a) Explain the different Types of Data Visualization and their applications. **[6]**
- b) Write a short note on Statistical Historiography in data visualization. **[6]**
- c) What are some key Scientific Design Choices in Data Visualization, and why are they important? **[6]**

- Q3)** a) Explain the differences between Combining and Merging Data Sets. When should each technique be used? **[6]**
- b) How does matplotlib compare to pandas and seaborn in terms of Data Visualization capabilities? **[6]**
- c) Describe the key steps involved in the process of data wrangling. **[5]**

OR

- Q4)** a) Describe the concept of Multidimensional Scaling and its role in Data Visualization. **[9]**
- b) What are Force-directed Techniques in Graph Layout, and how do they help in visualizing complex data structures? **[8]**

P.T.O.

- Q5) a)** Write a short note on [9]
- i) Data Aggregation
 - ii) Pivot Tables

- b)** Explain Cross Tabulation and its uses with suitable example. [9]

OR

- Q6) a)** What is the Group by mechanism in data aggregation, and how does it facilitate data analysis? [9]

- b)** What are Moving Window Functions, and how are they useful in smoothing and analyzing time series data? [9]

- Q7) a)** Discuss Data Reconstruction and its types in data modeling and visualization. [6]

- b)** Discuss a case study where medical image visualization and database modeling played a crucial role in diagnosis or treatment. [6]

- c)** Explain ultrasound images and its applications in data modeling and visualization. [5]

OR

- Q8) a)** Write a short note on [8]

- i) PET Images
- ii) Magnetic Resonance Images

- b)** How can tools like ER/Studio, Erwin Data Modeler, and DbSchema Pro aid in structuring and analyzing medical imaging data? [9]

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